

University Degree in Telematics Engineering
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Bachelor Thesis

"Machine Learning-Based Predictive Modeling of Energy Prices"

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SUMMARY

Keywords:

DEDICATION

Dedicated to my late grandfather who was not able to see me become an engineer like himself

Here is to my amazing family

To my parents, that have always To my grandparents

To my girlfriend

To my friends and colleagues

To anyone and everyone that has supported me during my years in university

Here is to the next steps in life

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1. INTRODUCTION TO THE PROBLEM

- **1. Explanation of how the price of energy works in Spain**
 - a. Energy
 - b. Prices
- **2. Machine Learning - Lets run through some model overviews, which are available to choose, but without actually referencing anything**

We initially discussed various ways to attack the problem at hand. We thought about different but settled on a plan to explore first established method of linear regression. This was a natural continuation of what was learned in the third year subject Modern Theory of Detection and Estimation.

Since the original ideal was to create an algorithm to predict hourly prices, we decided to simplify the problem. The first idea we developed was to separate out the project in 24 blocks, one for each hour block to predict. The reason behind splitting the data in 24 blocks was simple, we assumed that data between the same time slot across neighbor days, would be similar, or would follow a certain trend.

This kind of granular breaking up of data is similar to what a Random Forest does in order to absorb more details and create higher resolution predictions. We made this assumption based on the idea that it would simplify the project substantially.

2. BACKGROUND AND RELATED WORK

- **References**

- How is the price of electricity / energy set? What happens if new producers enter the market?
 - Articles that reference energy prices prediction
 - * As an inspiration for my own work:
 - 5-10 articles that do/talk about similar things
 - Energy consumption predictions
 - Energy prices in other European countries (Italy, Germany, UK, France)
 - Longer term predictions
 - * Predicciones de produccion
 - Describir tecnologias – explicar mas en detalle como funciona y enlaces a donde pueda averiguar el contenido
- Referenciar que metricas uso para medir mi error etc
 - Referencias de donde saco los datos
 - Referencias a que software utilizo referencias a documentacion de sklearn etc

3. PROPOSAL - THEORETICAL SYSTEM DESCRIPTION

Contar todos los pasos del pipeline de datos

Descarga de datos

Organización de datos

Limpieza de datos

Contar los bloques empleados en cada parte – como junto las piezas para construir el sistema – porque uso las piezas que uso

Hablar de implementaciones como que librerías, que scripts etc – no es muy necesario, mas importante la idea

This chapter will outline the theoretical design of the system developed for the project. It will outline the complete pipeline constructed for the system: starting from the data preparation stage, continuing with the data preparation to the final results.

Data preparation

Matrix creation

Mini-models - sliding window aproach

Technical indicators

ML blocks Linear Regression Linear combination of every feature Lasso Regression
Can turn unimportant features into 0s Random Forest Feature engineering

Pandas, NumPy, scikit-learn

4. EXPERIMENTATION

a. Data Description – datos del OMIE

b. Set Up Explanation – How do I use the previously explained system - what parameters did I modify to obtain the final results. Feature engineering, Alpha variation testing in Lasso, Tree depth and como lo usas en el sentido de tocar cosas, parametros como numero de arboles profundidad, el Alpha del lasso

c. Results a modo tabla unica comparando todas las iteraciones de cada tecnologia, y luego una conjunta con la mejor de todo (Si me hace falta mas, pues al apendice)

d. Low level discussion – conclusión que le contaria a un compañero de como he hecho todo esto

5. REGULATORY FRAMEWORK

6. SOCIO-ECONOMIC ENVIRONMENT

- a. Energy prediction
- b. Why is this relevant
- c. Shutdowns

7. CONCLUSIONS

- a. Recap
- b. Revisit Objectives
- c. Future Work